

SUPPLEMENTAL DEVICES

The following devices are supplemental and are compatible with the 4289.

4702 — Erasable and Electrically Reprogrammable ROM

- 2048 bits, organized 256 x 8
- Fast programming — 2 minutes for 2048 bits
- Inputs and outputs TTL compatible
- Three-state outputs
- Alterable Program Memory in system development

4316 — Mask Programmable ROM

- 16,384 bits, organized 2048 x 8
- Fully decoded

4101 RAM

- 1024 bits — organized 256 x 4
- Static operation
- Fully decoded
- Directly TTL compatible
- Three-state outputs
- Used for writeable Program Memory

3216/3226 — 4 bit Bidirectional Bus

- Low input load current
- High output drive
- Three-state outputs
- Inverted (3226) and non-inverted (3216) operations

THE FUNCTIONS OF A COMPUTER

This section introduces certain basic computer concepts. It provides background information and definitions which will be useful in later sections of this manual. Those already familiar with computers may skip this material, at their option.

A Typical Computer System

A typical digital computer consists of:

- a. A central processor unit (CPU)
- b. A memory
- c. Input/Output (I/O) ports

The program memory serves primarily as a place to store *instructions*, the coded pieces of data that direct the activities of the CPU. A group of logically related instructions stored in memory is referred to as a *program*. The CPU “reads” each instruction from memory in a logically determinate sequence, and uses it to initiate processing actions. If the program structure is coherent and logical, processing produces intelligible and useful results.

The data memory is used to store the data to be manipulated. The CPU can access any data stored in memory; but often the memory is not large enough to store the entire data bank required for a particular application. The program can be resolved by providing the computer with one or more *input ports*. The CPU can address these ports and input the data contained there. The addition of input ports enables the

computer to receive information from external equipment (such as a paper tape reader) at high rates of speed and in large volumes.

Almost any computer requires one or more *output ports* that permit the CPU to communicate the result of its processing to the outside world. The output may go to a display, for use by a human operator, to a peripheral device that produces “hard-copy”, such as a line-printer, to a peripheral storage device, such as a magnetic tape unit, or the output may constitute process control signals that direct the operations of another system, such as an automated assembly line. Like input ports, output ports are addressable. The input and output ports together permit the processor to interact with the outside world.

The CPU unifies the system. It controls the functions performed by the other components. The CPU must be able to fetch instructions from memory, decode their binary contents and execute them. It must also be able to reference memory and I/O ports as necessary in the execution of instructions. In addition, the CPU should be able to recognize and respond to certain external control signals, such as INTERRUPT and STOP requests. The functional units within a CPU that enable it to perform these functions are described below.

The Architecture of a CPU

A typical central processor unit (CPU) consists of the following interconnected functional units:

1. Registers
2. Arithmetic/Logic Unit (ALU)
3. Control Circuitry

Registers are temporary storage units within the CPU. Some registers, such as the program counter and instruction register, have dedicated uses. Other registers, such as the accumulator, are for more general purpose use.

ACCUMULATOR:

The accumulator usually stores one of the operands to be manipulated by the ALU. A typical instruction might direct the ALU to add the contents of some other register to the contents of the accumulator and store the result in the accumulator itself. In general, the accumulator is both a source (operand) and destination (result) register.

Often a CPU will include a number of additional general purpose registers that can be used to store operands or intermediate “scratch-pad” data.

PROGRAM COUNTER (JUMPS, SUBROUTINES AND THE STACK):

The instructions that make up a program are stored in the system's memory. The central processor examines the contents of the memory, in order to determine what action is appropriate. This means that the processor must know which location contains the next instruction.

Each of the locations in memory is numbered, to distinguish it from all other locations in memory. The number which identifies a memory location is called its *address*.

The processor maintains a counter which contains the address of the next program instruction. This register is called